

Multiple Choice (10 marks)

1. Calculate ΔG for the isothermal expansion of 2.25 mol of an ideal gas at 325 K from an initial pressure of 12.0 bar to a final pressure of 2.5 bar.
 - (a) $-7.89 \times 10^3 \text{ J}$
 - (b) $-12.75 \times 10^3 \text{ J}$
 - (c) $-8.01 \times 10^3 \text{ J}$
 - (d) $-10.02 \times 10^3 \text{ J}$
 - (e) $-11.32 \times 10^3 \text{ J}$
2. Why are there no impact craters on the surface of Io?
 - (a) Any craters that existed have been eroded through the strong winds on Io's surface.
 - (b) Io is protected by a layer of gas that prevents any objects from reaching its surface.
 - (c) The gravitational pull of nearby moons prevent objects from hitting Io.
 - (d) Io's magnetic field deflects incoming asteroids, preventing impact craters.
 - (e) Io did have impact craters but they have all been buried in lava flows.
3. An elevator is rising. In order to determine the tension in the cables pulling it, which values (for the elevator) would you need to know?
 - (a) Mass, velocity, and acceleration
 - (b) Mass, velocity, height, and acceleration
 - (c) Mass and velocity
 - (d) Velocity, height, and acceleration
 - (e) Velocity and acceleration
4. Say the pupil of your eye has a diameter of 5 mm and you have a telescope with an aperture of 50 cm. How much more light can the telescope gather than your eye?
 - (a) 1000 times more
 - (b) 50 times more
 - (c) 5000 times more
 - (d) 500 times more
 - (e) 10000 times more
5. When an element ejects a beta particle, the atomic number of that element

- (a) reduces by 1
- (b) increases by 3
- (c) increases by 1
- (d) reduces by half
- (e) increases by 4

True / False (10 marks)

Answer True or False

6. True or False: Terrestrial radiation has lower polarization and frequency compared to radiation from the Sun.
7. True or False: The proximity of a planet to other celestial bodies is the primary factor determining its volcanic and tectonic history.
8. True or False: The Fermi energy for copper is typically around 4.56 eV.
9. True or False: The power of the lens is calculated to be 3 diopters based on the given object and image distances.
10. True or False: Pluto has one medium-sized satellite and two small satellites orbiting it.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the magnitude of the gravitational force acting on a spherical water drop with a diameter of 1.20 m suspended in a downward-directed electric field of 462 N/C. Discuss the implications of your findings.
12. A free particle with a mass of 20 grams is initially at rest. Analyze the effects of applying a 100 dyne force for 10 seconds on its kinetic energy, detailing your calculations and underlying principles.

End of Paper